



American International University-Bangladesh (AIUB)
Faculty of Engineering
Department of CSE, EEE, and CoE

EEE4103 MICROPROCESSOR AND EMBEDDED SYSTEM
COURSE CAPSTONE PROJECT PROPOSAL FORM

SEMESTER: SUMMER 2025-26

PROJECT TITLE:

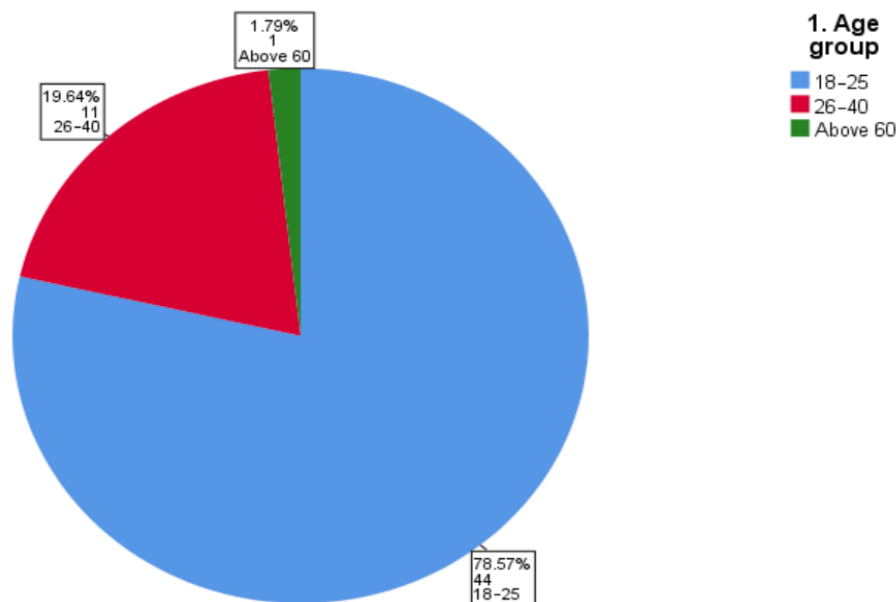
Smart Flood Monitoring and Early Warning System Using Arduino

SURVEY:

1. What is your age group?

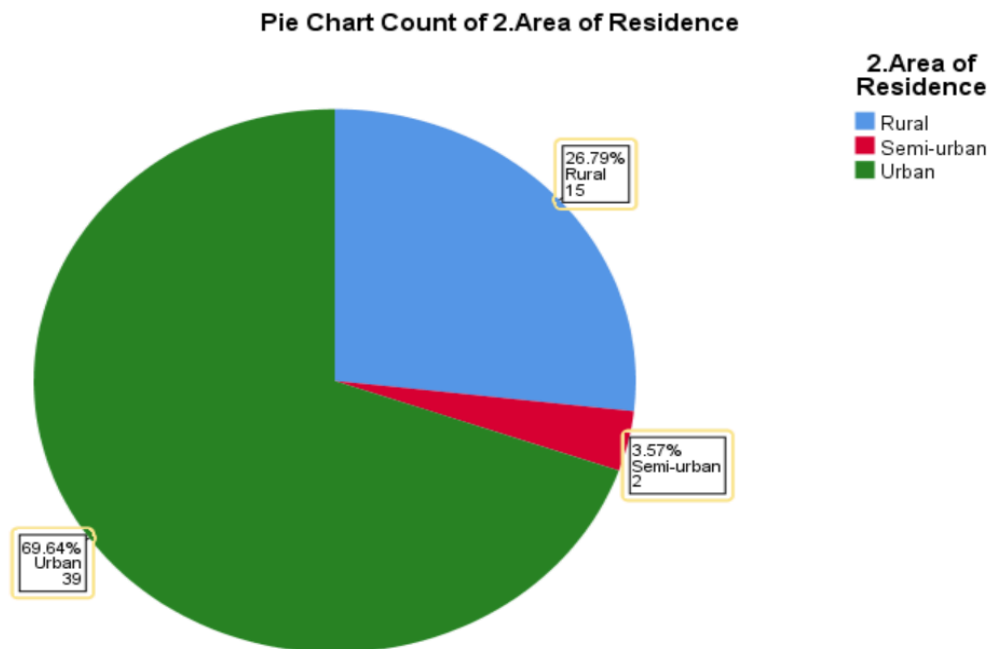
56 responses

Pie Chart Count of 1. Age group



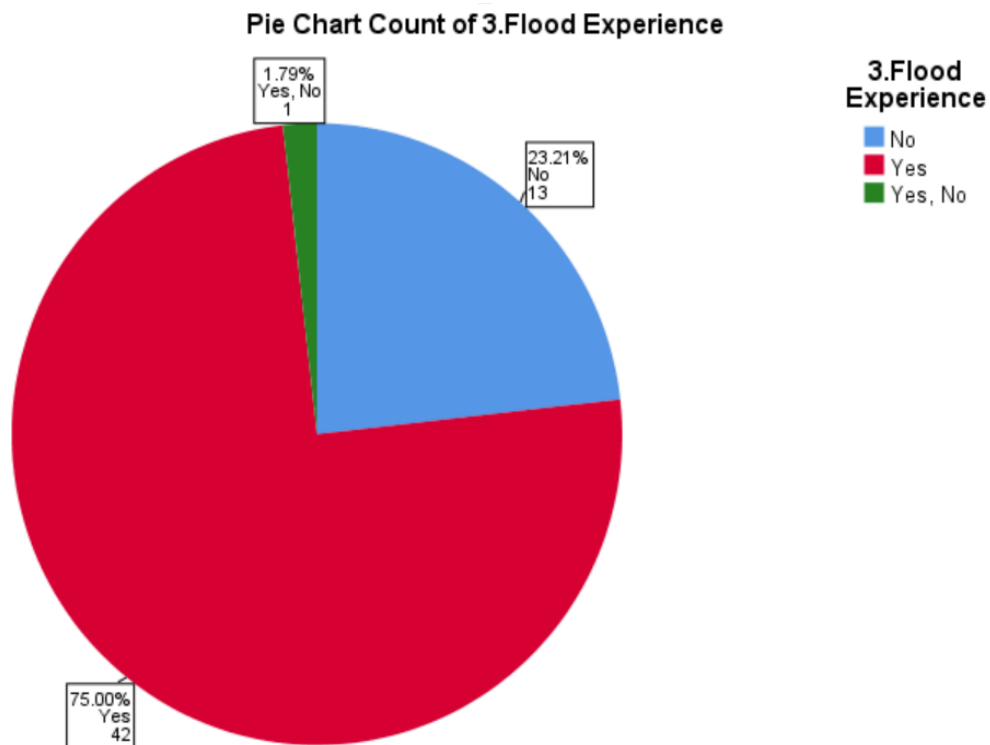
2. Which area do you currently live in?

56 responses



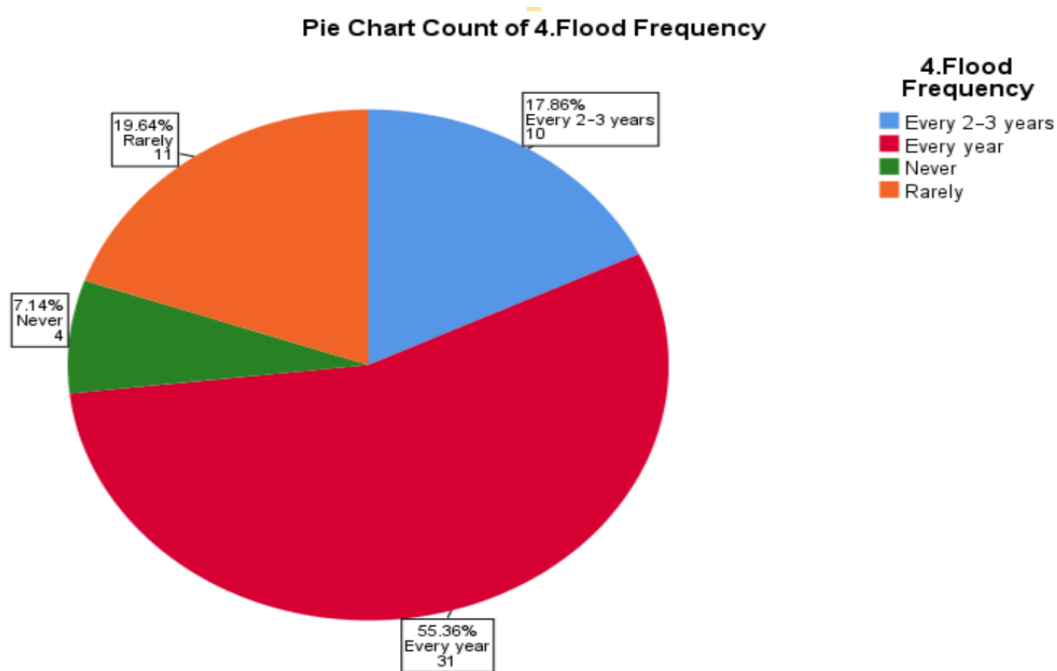
3. Have you ever experienced flooding in your area?

56 responses



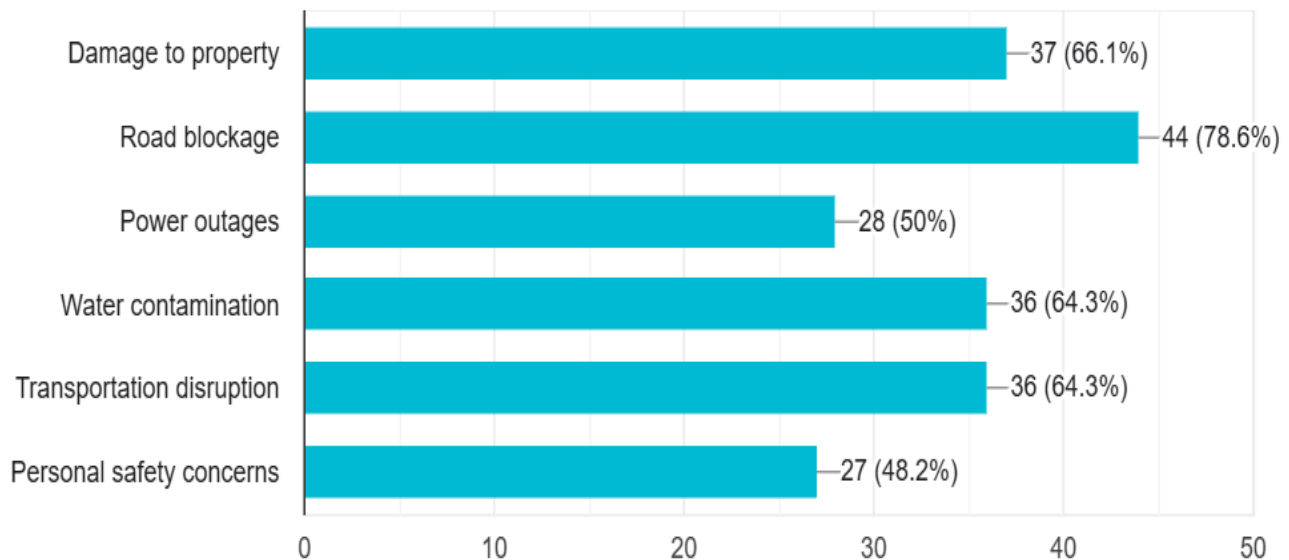
4. How often does flooding occur in your area?

56 responses



5. Which problems do you usually face during flooding? (Multiple answers can be selected)

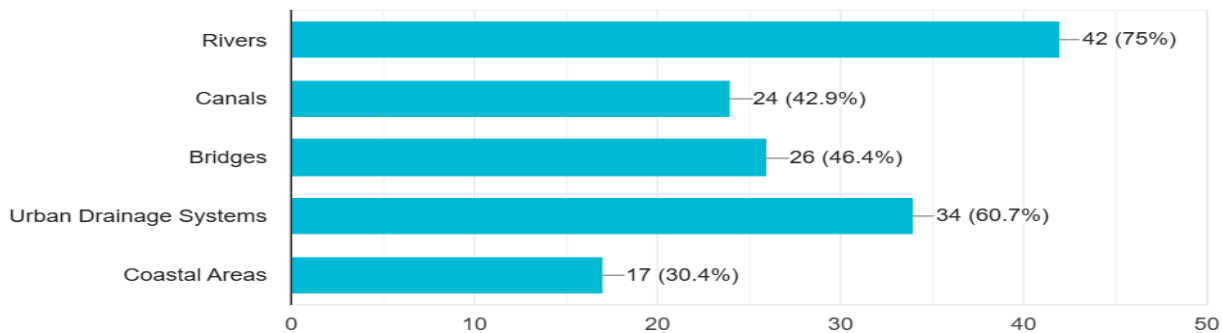
56 responses



6. Which place should be given priority for installing flood monitoring devices?

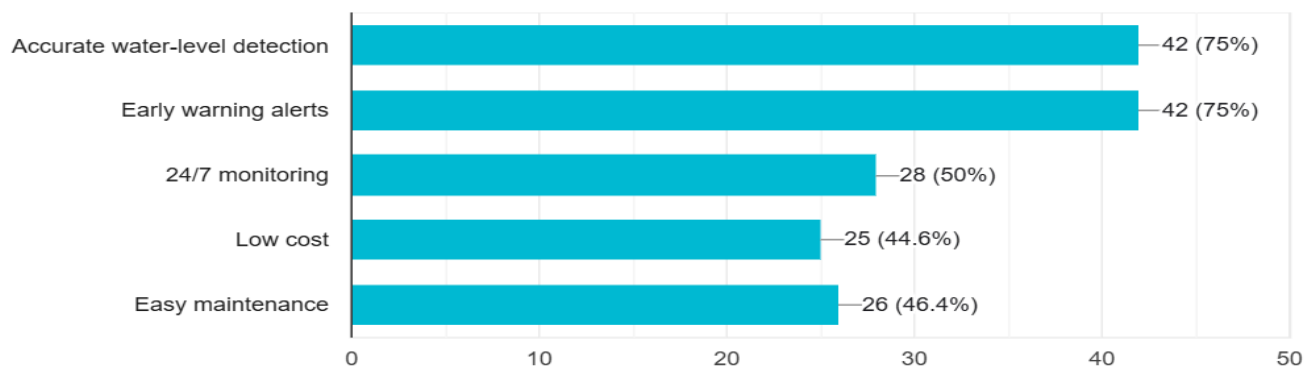
(Multiple answers can be selected)

56 responses



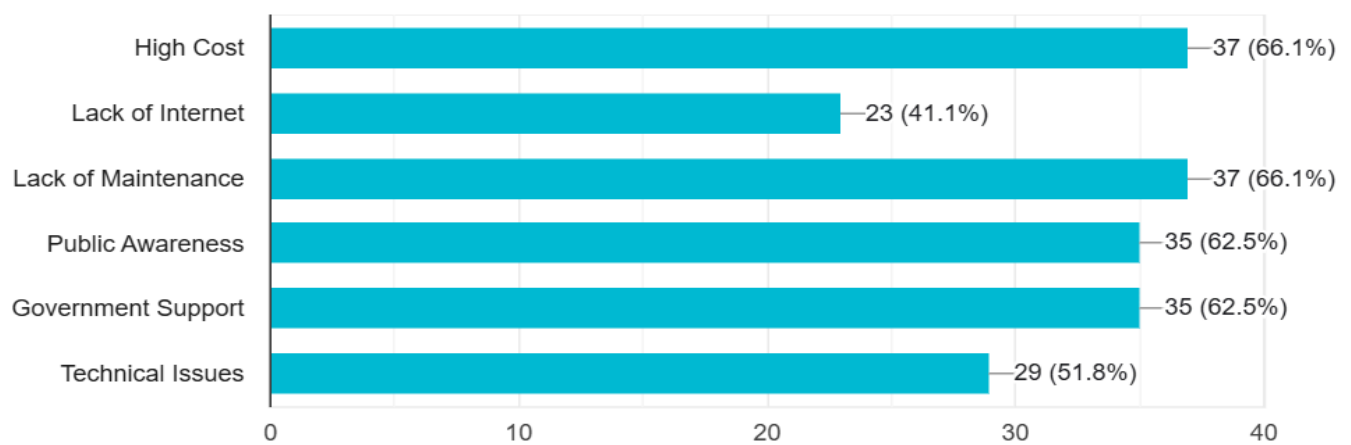
7. What feature do you think is most important in a smart flood monitoring system? (Multiple answers can be selected)

56 responses



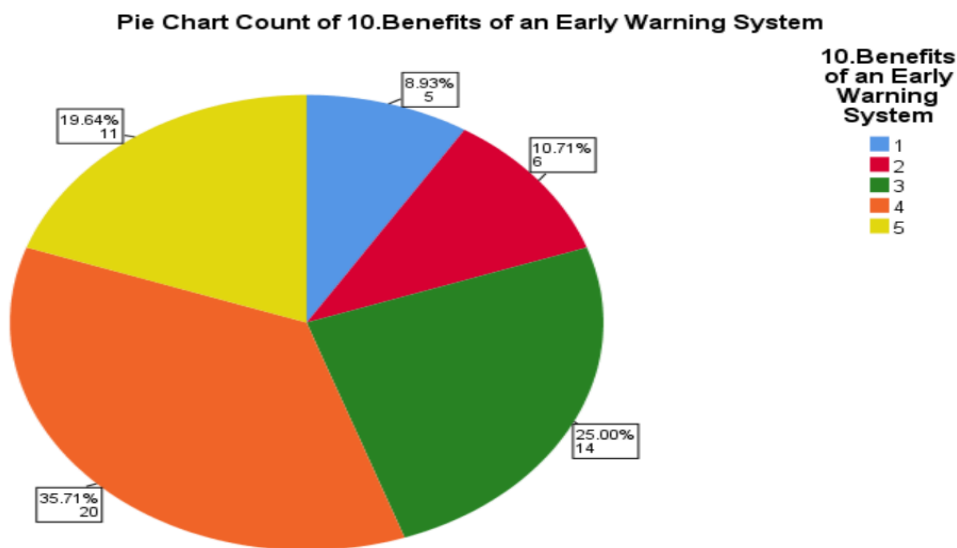
8. What is the biggest challenge in implementing a smart flood monitoring system? (Multiple answers can be selected)

56 responses



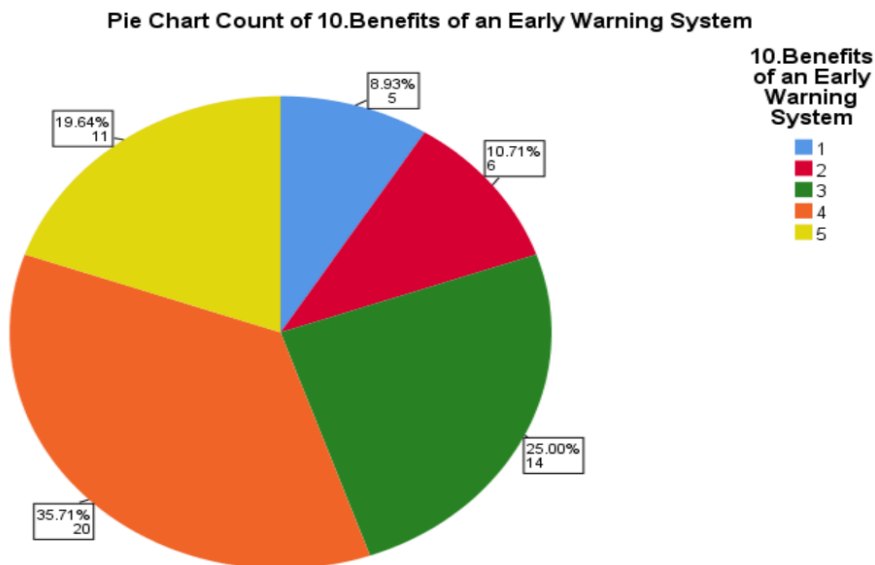
9. How satisfied are you with the current flood warning system?

56 responses



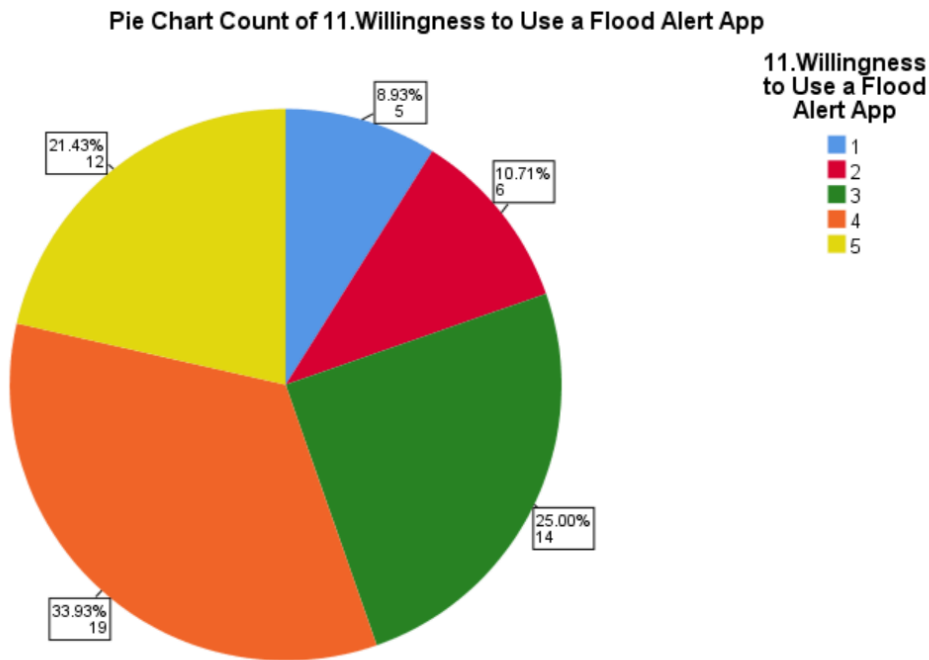
10. Do you think an early warning system can reduce flood-related casualties and property damage?

56 responses



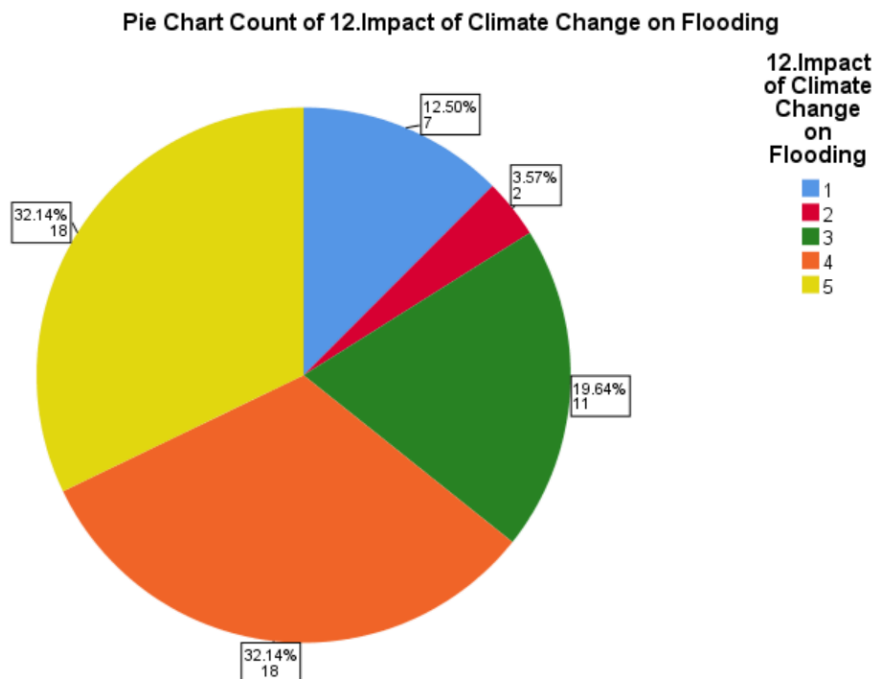
11. Would you use a mobile application that provides real-time flood alerts?

56 responses



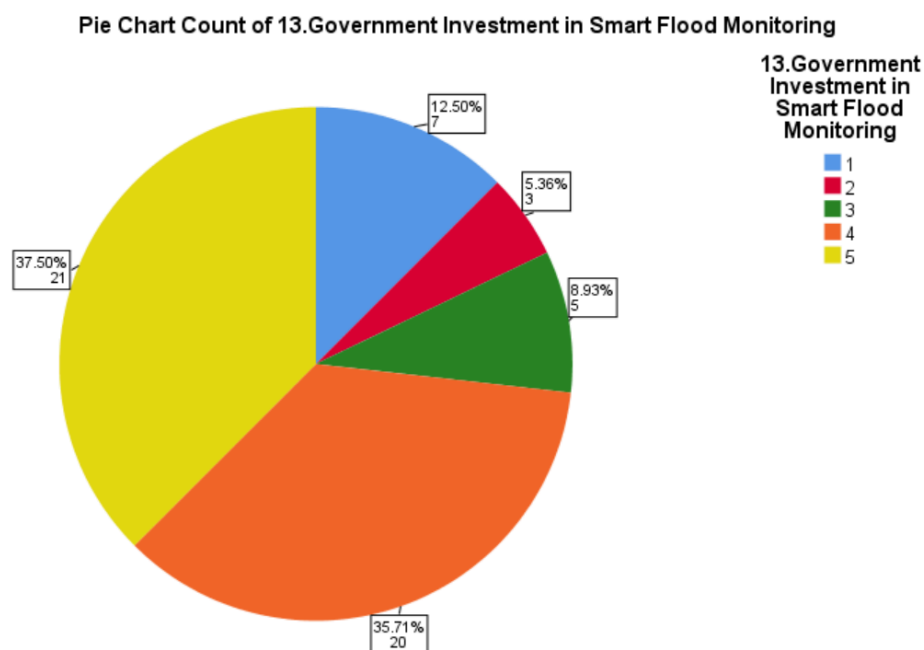
12. Do you believe climate change has increased flooding in Bangladesh?

56 responses



13. The government should invest more in smart flood monitoring technologies.

56 responses



SurveyLink:

https://docs.google.com/forms/d/e/1FAIpOLSeKW_shc-UKJd9XBWSJChJlbNP9r1fXGxqU1IZRBVecvOAKZg/viewform?usp=header

AIMS AND OBJECTIVES OF THE PROJECT:

The primary aim of this project is to design a low-cost, embedded flood monitoring and early warning system using Arduino integrated with ultrasonic water level sensors, rainfall sensors, and a GSM/GPRS module for real-time data transmission.

To implement an early warning mechanism that triggers automated SMS/buzzer/LED alerts to nearby residents and local authorities when water levels cross a predefined critical threshold.

To simulate the complete sensor-to-alert data flow using Proteus to validate system response time and reliability before hardware deployment.

To analyze the collected water level and rainfall data through threshold-based reasoning (or a basic prediction algorithm) to differentiate between normal, warning, and critical flood levels.

To evaluate the system's accuracy and response time under varying simulated flood scenarios to ensure dependable real-world performance.

LITERATURE REVIEW:

Flood-related disasters continue to be a major hazard to human existence and the economy across the globe, making it imperative that there should be an automation of the process by use of technology. Recent research explores the implementation of the Internet of Things (IoT) and wireless sensor networks to develop efficient Early Warning Systems (EWS) that provide vulnerable communities with localized data and critical lead times.

Ganancias [1] developed a microcontroller-based EWS specifically for remote, high-risk regions, uniquely adopting a multi-hazard approach to monitor both flood levels and landslide risks through water level, soil moisture, and ground displacement sensors. Methodologically, the study utilized a descriptive-evaluative design and integrated environmental sensors with a GSM module to ensure offline functionality, addressing the instability of internet connectivity in rural disaster zones. The system demonstrated high compliance with ISO 25010 quality standards; however, the researcher noted that current institutional responses remain largely reactive due to incomplete geographic coverage and limited equipment availability at the local level.

Unlike the institutional focus of Ganancias, Ayorinde et al. [2] prioritized a community-level implementation in Kroo Bay, Sierra Leone, specifically addressing the physical security of the hardware. The methodology employed an Arduino Uno with an HC-SR04 ultrasonic sensor and a SIM800L GSM module for SMS alerts, achieving a measured accuracy of ± 1.8 cm and an SMS latency of 5–10 seconds. Unlike others who only concentrated on technical aspects, Ayorinde et al. addressed real-world social challenges by redesigning the system casing to be cemented into walls after initial units were stolen. Despite its success, the study highlighted GSM network failure during extreme rainfall and a lack of long-term testing during dry seasons as significant limitations.

Ramadan and Sulistya [3] proposed a low-cost IoT solution but, in contrast to the non-contact ultrasonic systems of Ayorinde et al. or Ganancias, utilized a conductive K-0135 water level sensor. The primary objective of this research was to eliminate false alarms by the use of threshold detection in a static as well as in a dynamic environment like raindrops. Using a NodeMCU ESP8266 and HTTP-based communication, they achieved a 100% communication success rate in laboratory settings. The problem with this design is that it has been developed and tested only within a closed laboratory setup without considering real-world conditions such as accumulation of debris, sediment buildup, or any degradation in the sensors.

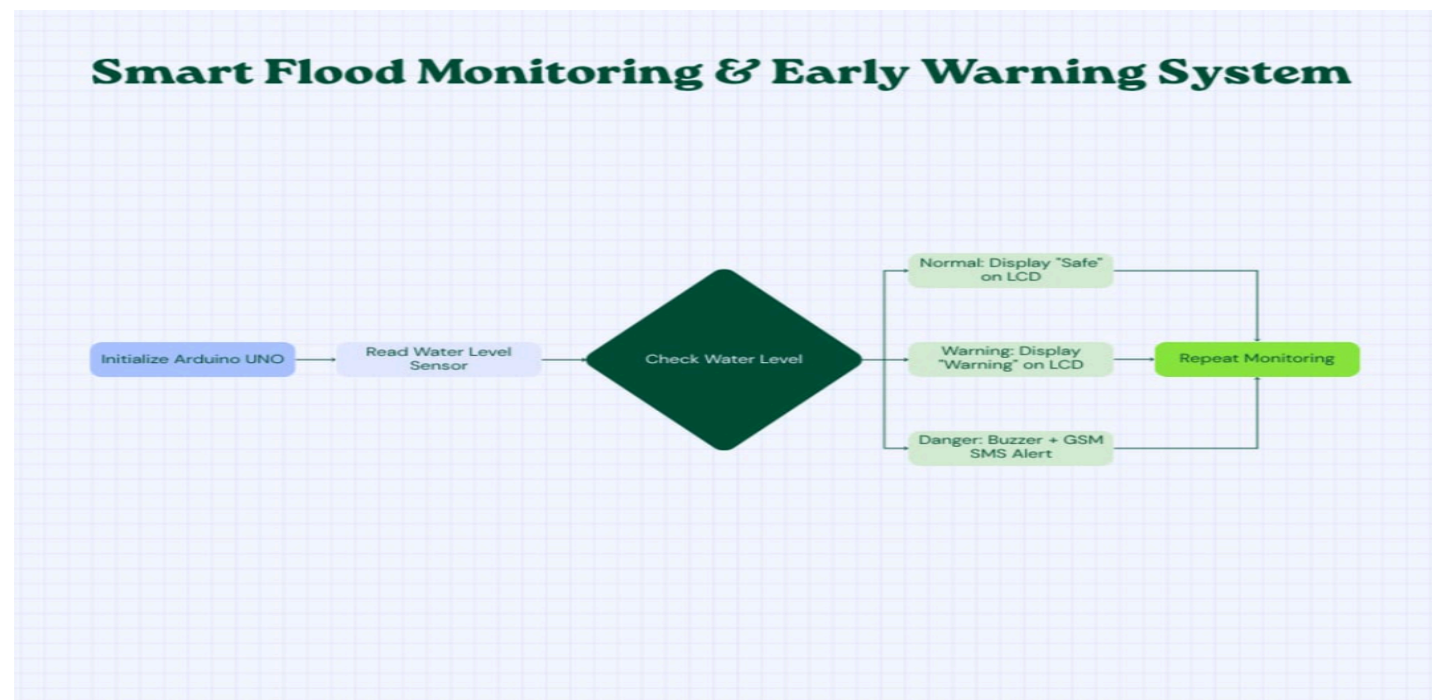
Faturochman et al. [4] returned to ultrasonic sensing, developing an ESP32-based monitoring system integrated with a Telegram chatbot for real-time public alerts. Following the waterfall development method, the researchers successfully demonstrated that sensor-calculated water levels matched conventional manual measurements exactly across seven experiments. While this study prioritized public accessibility through social media, it was limited to simple threshold-based logic (Siaga 1-3) and, unlike the work of Mansuit et al., lacked integrated atmospheric data to provide a holistic view of a storm's progression.

Susanti et al. [5] specialized the application of ultrasonic sensing for tidal flood monitoring in Semarang, Indonesia. Using an ESP32S2 microcontroller and a high-precision A01NYUB ultrasonic sensor, the methodology focused on optimizing tidal data with an accuracy characterized by an RMSE of 0.08943. Compared to the river-based monitoring of Ramadan and Faturochman, this research specifically addressed coastal resilience. Nevertheless, the researchers acknowledged that the absence of digital signal filtering

necessitated physical distancing (166 cm from the tower) to prevent the sensor from capturing wave noise rather than the mean sea level.

Mansuit et al. [6] expanded the sensing scope by integrating water level measurement with atmospheric parameters such as temperature, humidity, and barometric pressure using AHT20 and BMP280 sensors. Their research process entailed the use of an adaptive sampling technique which varied according to the weather, analyzed using a Django-powered website application. Addressing the localized constraints found in Ayorinde et al., Mansuit et al. proposed a scalable cloud-based framework; however, its current evaluation was limited strictly to simulated computer-generated data, meaning the hardware's actual field performance and accuracy remain unverified.

EXPERIMENTAL BLOCK DIAGRAM:



POSSIBLE OUTCOMES OF THE PROJECT:

1. Alerts the people of those areas which are prone to floods regarding the approaching danger so that they have enough time to leave the area or save themselves.
2. Helps the local authorities and disaster management agencies take quick decisions based on the data regarding water levels available in real time.
3. Decreases financial damage in agriculture and small-scale enterprises by helping communities take precautionary steps before flooding takes place.
4. Encourages the inhabitants of flood-prone zones to be ready for disasters and move from being reactive to proactive when it comes to flooding.
5. Offers a low-cost, scalable model that can be replicated across rural and underserved regions where expensive commercial flood-monitoring infrastructure is not accessible.
6. Increases environmental consciousness among communities, enabling them to deal with changes in rainfall and river levels due to climate

PROJECT TIMELINE (GANTT CHART):

PROJECT TIMELINE (GANTT CHART)																									
PROJECT TITLE: Smart Flood Monitoring and Early Warning System																									
ID	TASK / ACTIVITY	PHASE	START WEEK	DURATION (WEEKS)	1	2	3	4	5	6	WEEKS								7	8	9	10	12	13	14
1	Project Selection & Team Meeting	Initiation	1	1																					
2	Problem Identification & Need Analysis		1	2																					
3	Literature Review		1	3																					
4	Requirement Analysis		2	2																					
5	Survey Design & Questionnaire Preparation		2	2																					
6	Survey Execution & Data Collection		3	2																					
7	Survey Results Analysis & Outcome Summary		5	2																					
8	Define Project Aims & Objectives	System Design	3	1																					
9	System Architecture & Block Diagram		3	2																					
10	Component Selection & Specification		4	2																					
11	Circuit Design		5	2																					
12	Software Design (Flowchart & Algorithm)	5	2																						
13	Simulation Setup	Simulation & Testing	6	1																					
14	Simulation & Functional Testing		6	1																					
15	Hardware Procurement	Implementation (Hardware)	6	1																					
16	Project Proposal Preparation		6	1																					
17	Review & Finalization of Proposal		7	1																					
18	Submission of Project Proposal	7	1																						
19	Hardware Assembly & Integration	Implementation (Execution)	-	-																					
20	Firmware Coding & Development		-	-																					
21	System Testing & Debugging	Testing & Evaluation	-	-																					
22	Performance Evaluation & Data Analysis		-	-																					
23	Report Writing & Documentation	Documentation & Finalization	-	-																					
24	Final Presentation & Report Submission		-	-																					

REFERENCES:

- [1] F. M. Ganancias, "Microcontroller-Based Early Warning System for Floods and Landslide Prone Areas," *The Light Explorer*, vol. 10, no. 1.4, Mar. 2025, doi: 10.57180/jmhd8825.
- [2] O. B. Ayorinde, S. F. Kamara, E. Fullah, and A. J. Conteh, "Performance Evaluation of Arduino-Based Ultrasonic Sensors and GSM SMS for Community-Level Flood Warnings," *Journal of Computer Science and Technological Enquiry (JCSTE)*, vol. 2, no. 1, pp. 35–53, May 2026, doi: 10.65496/jcste.2026.74.
- [3] G. Y. A. Ramadan and E. Sulistya, "Design and Evaluation of an IoT-Based Flood Early Warning System Using Conductive Water Level Sensor," *Digital System and Computing*, vol. 2, no. 1, pp. 1–8, May 2026, doi: 10.58920/dsc0201572.
- [4] W. Faturachman, P. Dauni, E. T. A, A. Husain, and I. Abdurrohm, "Design of Flood Early Warning Monitoring System Using a ESP32 Microcontroller Ultrasonic Sensor Based on Website and Telegram Chat Bot," *Journal Elektronik Sistem Informasi (JESII)*, vol. 1, no. 2, pp. 101–110, Dec. 2023.
- [5] A. F. S. Susanti *et al.*, "Instrumentation for Monitoring and Early Warning of Tidal Flood Using ESP32S2 Microcontroller and A01NYUB Ultrasonic Sensor in Tambakrejo, Semarang," *IOP Conference Series: Earth and Environmental Science*, vol. 1350, no. 1, p. 012046, 2024, doi: 10.1088/1755-1315/1350/1/012046.
- [6] S. Mansuit, N. bin Abdul Taha, A. Bono, and J. Sulindap, "Design of Web-Based Flood Monitoring System Using ESP32 Microcontroller: An Integrated IoT Solution for Early Warning," *IOP Conference Series: Earth and Environmental Science*, vol. 1605, no. 1, p. 012025, 2026, doi: 10.1088/1755-1315/1605/1/012025.

Instructions:

1. There is **no definite format for writing the course project proposal**, but students must follow the instructions mentioned above properly.
2. Fill in the form accurately with all necessary information.
3. Make a color print on this form.
4. Figures, tables, charts, circuit diagrams, block diagrams, and wave shapes must be color printed.
5. The survey form links with the answers that must be provided in the proposal form

FOR FACULTY USE ONLY**COMMENTS BY COURSE TEACHER:**

COURSE TEACHER'S NAME**COURSE TEACHER'S SIGNATURE****DATE****GROUP MEMBERS**

(Maximum **6 students** are permitted to carry out a single Project. However, depending on the capability of the students, **4 students** may be allowed, but not fewer than that.

NAME: Tajmun Nahar Tisha ID #: 23-54386-3 PROGRAM: CSE EMAIL: 23-54386-3@student.aiub.edu	NAME: Saharia Sanzida Liya ID #: 23-53055-3 PROGRAM: CSE EMAIL: 23-53055-3@student.aiub.edu
NAME: Zarin Anan ID #: 23-55874-3 PROGRAM: CSE EMAIL: 23-55874-3@student.aiub.edu	NAME: Lubna Islam Shuzona ID #: 24-56570-1 PROGRAM: CSE EMAIL: 24-56570-1@student.aiub.edu
NAME: Kazi Ayesha Afia Rayeesa ID #: 23-52794-2 PROGRAM: CSE EMAIL: 23-52794-2@student.aiub.edu	NAME: ID #: PROGRAM: EEE/CoE/CSE EMAIL:

Assessment Materials and Marks Allocation:

CO	Assessment Materials	POIs	Marks
CO2	Course Capstone Proposal Form	P.c.2.C6 P.c.3.C5	50

Course Name:	Microprocessor and Embedded Systems Lab	Course Code:	EEE 4103
Semester:	Summer 2025-26	Sec:	U
Faculty Member:	DEBRAJ DAS		

Capstone Project Title:	Smart Flood Monitoring and Early Warning System Using Arduino		
Project Group No.	06		
Sl #	Student ID #	Student Name	Obtained Marks
1.	23-54386-3	Tajmun Nahar Tisha	
2.	23-53055-3	Saharia Sanzida Liya	
3.	23-55874-3	Zarin Anan	
4.	24-56570-1	Lubna Islam Shuzona	
5.	23-52794-2	Kazi Ayesha Afia Rayeesa	

Assessed POI:

CO/ CLO Number	CO/CLO Statement	K	P	A	Assessed Program Outcome Indicator	BNQF Indicator	Teaching-L earning Strategy	Assessment Strategy
2	Develop a survey to implement a capstone project considering cultural and societal concerns.	K5	P1, P2-3 P6-7		P.c.2.C6 P.c.3.C5	PS.2	Discussion	Course Project Proposal Form

Assessment Rubrics:

KPIs	Excellent [2]	Proficient [1.5]	Good [1]	Acceptable [0.5]	Unacceptable [0]	No Response [0]	Secured Marks
Project Title	The title reflects an issue related to complex engineering problems, showing targets and methods with possible outcomes.	The title reflects an issue related to complex engineering problems, showing targets and methods, but some missing issues.	The title reflects an issue related to the course capstone project, but there may be some missing issues.	The title reflects an issue related to the course capstone project, but it is not complete or specific.	The title does not reflect any issues related to the course capstone project.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (2)	

KPIs	Excellent [10-12]	Proficient [7-9]	Good [4-6]	Acceptable [2-3]	Unacceptable [1]	No Response [0]	Secured Marks
Survey	The survey developed as a process for complex engineering problems, considering cultural and societal factors, has superior variables, targets, measures, and the implementation process is clear and challenging for future project implementation, with several possible outcomes having good impacts.	The survey, developed as a process for complex engineering problems, considering cultural and societal factors, has good variables, targets, measures, and the implementation process is clear and challenging for future project implementation, with some possible outcomes and little impact.	The survey developed as a process for complex engineering problems, considering cultural and societal factors, has moderate variables, targets, measures, and The implementation process is clear and challenging for future project implementation, with a few possible outcomes and impacts.	The survey, developed as a process for complex engineering problems, considering cultural and societal factors, has good variables, targets, measures, and the implementation process is somewhat clear for future project implementation, with very few possible outcomes and little impact.	The survey developed as a process for complex engineering problems, considering cultural and societal factors, has poor variables, targets, measures, and the implementation process is very unclear for future project implementation, with few possible outcomes but no impacts.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (12)	

KPIs	Excellent [4]	Proficient [3]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
Aims and Objectives	Aims and objectives are written to solve complex engineering problems, considering cultural and societal factors with specific targets, measurements, and implementation processes that are clear and challenging, and have several possible outcomes having very good impacts.	Aims and objectives are written to solve complex engineering problems, considering cultural and societal factors with general targets, measurement, and implementation processes that are not clear and challenging, and have some possible outcomes that have a good impact.	Aims and objectives are written to solve complex engineering problems considering a few cultural and societal factors with narrow targets; measurement, and implementation processes are clear and challenging and have a few possible outcomes, having some impacts.	Aims and objectives are written to solve complex engineering problems considering cultural or societal factors with a very specific target; measurement and implementation processes are not clear or challenging, and have little possible outcome, having no impact.	Aims and objectives are written to solve complex engineering problems, but do not consider cultural and societal factors with any targets; measurement and implementation processes are not clear and challenging, and no possible outcomes that have an impact.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (4)	

KPIs	Excellent [10-12]	Proficient [7-9]	Good [4-6]	Acceptable [2-3]	Unacceptable [1]	No Response [0]	Secured Marks
Literature Review	Specific formats are maintained to review and cite the literature with recent publications. Identified and analyzed the problem correctly.	Specific formats are maintained to review and cite the literature with recent publications. Identified and analyzed the problem correctly, but not all issues were addressed with relevant or intended work.	Specific formats are maintained to review and cite the literature with recent and past publications. Identified and analyzed the problem correctly, but not all issues were addressed with relevant or intended work.	Specific formats are maintained to review and cite the literature with recent and past publications. Identified but could not analyze all the problems correctly, and all issues were not addressed with relevant or intended work.	No specific formats are maintained to review and cite the literature with recent publications. Could not identify and analyze all the problems correctly, and all issues are not addressed with relevant or intended work at all.	No Response at all/ copied from others/ identical submissions with gross errors/ image file printed	
Comments						Total Marks (5)	

KPIs	Excellent [5-6]	Proficient [4]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Experimental Block Diagram	The block diagram is drawn to show the connections of all the possible components or subsystems to show their interdependence with all possible flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of all of the possible components or sub-systems to show their interdependence, with a few missing flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of most of the possible components or sub-systems to show their interdependence, with a few missing flows of signals from inputs to outputs.	The block diagram is drawn to show the connections of a few possible components or sub-systems to show their interdependence, with some missing flow of signals from inputs to outputs.	The block diagram is not drawn to show the connections of all possible components or sub-systems, to show their interdependence and flow of signals from inputs to outputs.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (5)	

KPIs	Excellent [4]	Proficient [3]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
Possible Outcomes	Outcomes are written to achieve complex engineering problems' solutions, considering cultural and societal factors and showing measurement and implementation processes to attain the outcomes with all possible impacts.	Outcomes are written to achieve complex engineering problems' solutions, considering cultural and societal factors and showing measurement and implementation processes to attain the outcomes with some impacts.	Outcomes are written to achieve complex engineering problems' solutions, considering cultural and societal factors, and do not show measurement and implementation processes to attain the outcomes without showing any impacts.	Outcomes are written to achieve complex engineering problems' solutions, but do not consider cultural and societal factors, and do not show measurement and implementation processes to attain the outcomes without showing any impacts.	Outcomes are not written to achieve complex engineering problems' solutions, do not consider cultural and societal factors, and do not show measurement and implementation processes to attain the outcomes without showing any impacts.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (4)	

KPIs	Excellent [6-7]	Proficient [4-5]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
Gantt Chart	Specific formats are maintained to draw the Gantt chart, and there is an order of workflow with all work to be done.	Specific formats are maintained to draw the Gantt chart, and there is an order of workflow with a few tasks missing.	Specific formats are maintained to draw the Gantt chart, and there is an order of workflow with some work missing.	No specific formats are maintained to draw the Gantt chart, and there is little order to the workflow, with some work missing.	No specific formats are maintained to draw the Gantt chart, and there is no order of workflow, with the most important tasks missing.	No Response at all/ copied from others/ identical submissions with gross errors/ image file printed	
Comments						Total Marks (5)	

KPIs	Excellent [3]	Proficient [2.5]	Good [2]	Acceptable [1]	Unacceptable [0.5]	No Response [0]	Secured Marks
References	Specific formats are maintained to write references, and all are recently published journal and conference papers having no missing information.	Specific formats are maintained to write the references, and all are journal and conference papers, but some old papers have missing information.	No specific formats are maintained to write the references, and many are internet sources with several missing information and very old references.	No specific formats are maintained to write the references, and most of them are internet sources with missing information.	No specific formats are maintained to write the references, and all are internet sources with missing information.	No Response at all/ copied from others /identical submissions with gross errors/ image file printed	
Comments						Total Marks (3)	